

CHROMA GUARD



“WE DON’T JUST CATCH
POLLUTION - WE STOP IT AT
THE PIPE.”

PRESENTED BY :
TEAM
CHROMA GUARD

PRESENTED AT :
HACK FUSION 2026

**EFFLUENT COMPLIANCE
SYSTEM**

PROBLEM

Textile dye effluent compliance is checked in a lab, days after the water has already left the pipe.

1.

REACTIVE, NOT REAL-TIME

Lab-based effluent testing takes days. By the time a violation is confirmed, the discharge has already reached the river or groundwater.

3.

HUNDREDS OF SMALL DYEING UNITS

Small and mid-sized units often can't afford full-scale ETPs, so partially-treated effluent is a routine, ongoing occurrence.

2.

NO AUTOMATIC INTERVENTION

Even where sensors exist, they log readings — they don't stop the flow. Detection without action doesn't prevent harm.

4.

INEFFICIENT OPERATIONS IRREVERSIBLE ECOLOGICAL COST

Dye effluent laden with TDS, chlorides, and heavy metals degrades rivers, groundwater, and farmland long before regulators can act.

CASE STUDY

Noyyal River, Tirupur – Erode Belt

India's textile-dyeing belt is right in this region — the case for ChromaGuard is local, not hypothetical.

700+

dyeing & bleaching units operate
along the Noyyal basin

~96 MLD

of wastewater generated by
Tirupur's dyeing units daily

18

Common Effluent Treatment Plants
(CETPs) currently running

3–4×

river TDS levels exceed the 2,100
mg/L discharge standard

What went wrong — and why detection alone hasn't fixed it

Untreated effluent flowed into the Noyyal River for decades before the Supreme Court intervened in 1996, ordering non-compliant units to shut down.

The Madras High Court later mandated zero-liquid-discharge treatment in 2011 — yet published water-quality studies still show river TDS several times above the permissible limit.

The gap isn't awareness or regulation — it's that compliance is checked periodically in a lab, while discharge happens continuously at the pipe. ChromaGuard closes exactly that gap.

CONCEPT ORIGIN

From “monitor” to “intervene”

The first version of this idea was a pollution dashboard — sensors reporting dye levels to an app. It was a good detector, but one question reframed it: a dashboard tells you pollution happened. It doesn’t stop it.

That reframing — detect, decide, and physically act on the same device — is what turned a monitoring gadget into ChromaGuard: a compliance system with a valve, not just a screen.

V1 - SENSING

Turbidity + pH + TDS logging to a cloud dashboard.

V2 – CLASSIFICATION

Rule-based fusion classifier (weighted thresholds across 4 sensors) today; a trained ML classifier is the planned upgrade once field-labeled dye samples are collected.

V3 – INTERCEPTION

Automated solenoid valve diverts non-compliant discharge; fails to “divert” by default.

V4 – TREATMENT

Dosing pump + holding tank as an on-site pre-treatment buffer, not a full ETP replacement.

Sense. Classify. Decide. Act.

ChromaGuard is a pipe-mounted sensor-and-actuator pod that detects, classifies, and physically intercepts non-compliant dye effluent in real time.



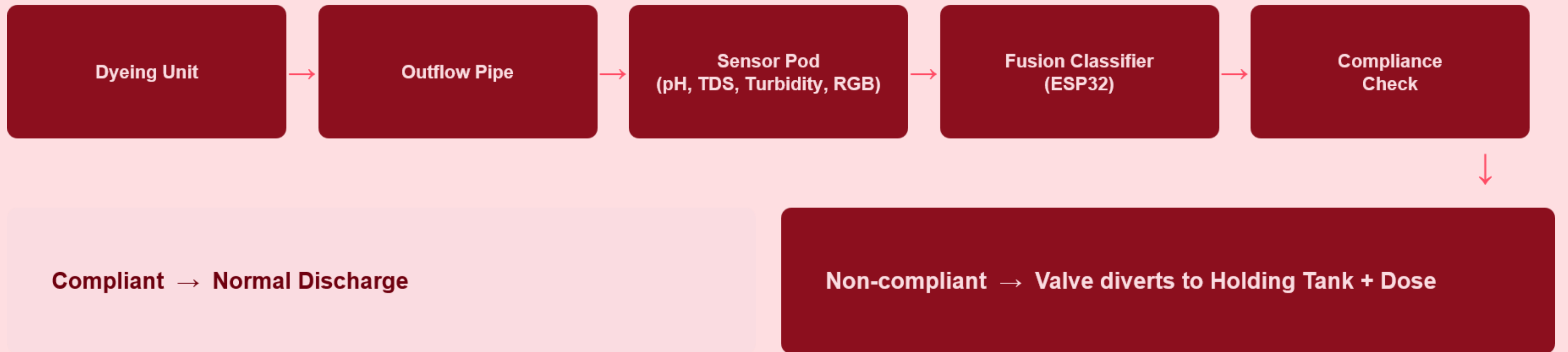
Where the ~5s comes from: Sense ~1.0s → Classify ~1.0s → Decide ~0.5s → Valve actuation ~1.5–2.0s

Optional Plan B — Treat

Diverted batches route into a small holding tank where a dosing pump releases a coagulant — a pre-treatment buffer stage ahead of the CETP, not a substitute for full effluent treatment.

CONCEPT DIAGRAM

CONTINUOUS SENSING → AUTOMATED ACTION



Fail-safe by design: the valve defaults to DIVERT on sensor fault or power loss. The operator retains a manual override switch, and every manual bypass is logged for auditability — addressing liability directly.

WHAT IT'S SOLVING

**FOUR GAPS,
CLOSED AT
THE PIPE**

THE DETECTION-TO-ACTION TIME GAP

Lab turnaround of days becomes an on-pipe decision in under 5 seconds.

IRREVERSIBLE DOWNSTREAM CONTAMINATION

Diversion happens before effluent reaches the river or groundwater — not after.

SILENT, UNWITNESSED DISCHARGE EVENTS

Continuous automated monitoring removes reliance on periodic inspection or self-reporting.

WEAK ENFORCEMENT VISIBILITY FOR REGULATORS

Every diversion event is logged, giving TNPCB an auditable, tamper-evident compliance trail.

WHY IT'S BETTER

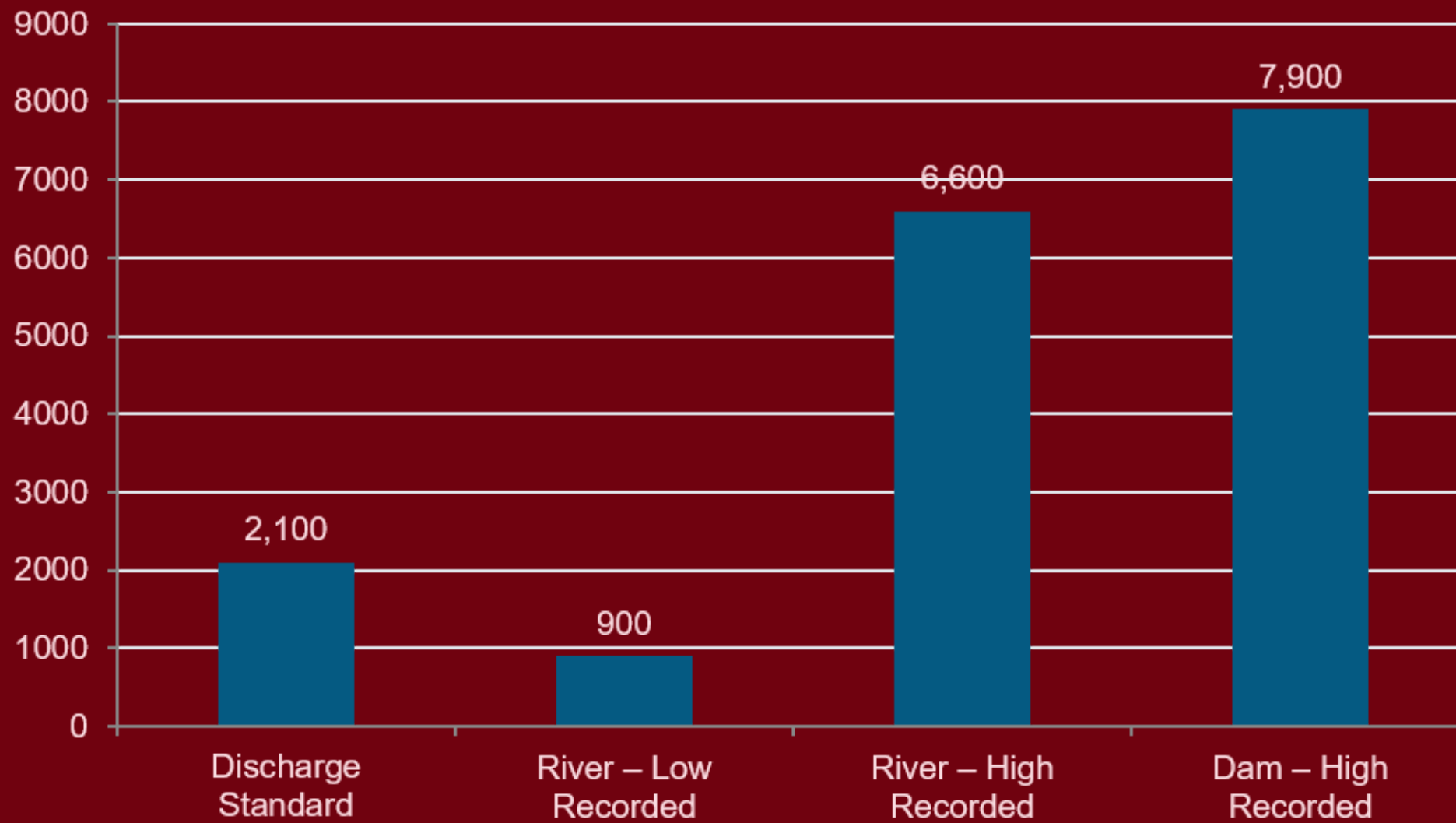
DETECTION ALONE WAS NEVER THE FINISH LINE

	Lab-based Testing	Passive IoT Monitors	ChromaGuard
Speed	Result in 2–5 days	Reading in real time, no action taken	Classified + acted on in <5 seconds
Response	Detects after discharge	Logs after discharge	Diverts before discharge
Reliability	Manual sampling, easily missed	Data-only, needs a human to respond	Fully automated intervention
Audit trail	No continuous audit trail	Dashboard logs only	Tamper-evident action log for regulators

ANALYTICAL DATA

Published water-quality studies on the Noyyal River basin - the direct evidence base for this project.

TDS Levels vs. Permissible Standard (mg/L)



3-4×

river TDS exceeds the discharge standard at peak points

18

CETPs operating, yet standards are still routinely exceeded

700+

units to eventually cover — pilots start with 1-2 CETP-linked units

Source: peer-reviewed water-quality studies of the Noyyal River basin, Tirupur – Erode region (TNPCB discharge standard: 2,100 mg/L TDS).

BUSINESS MODEL

A B2G + B2B MODEL: SELL COMPLIANCE INFRASTRUCTURE TO REGULATORS, DEPLOY IT AT THE FACTORIES THEY REGULATE.

01

Hardware Sale / Lease

Sensor pod + valve unit sold to CETPs and large dyeing units, or leased to lower entry cost for small units.

02

SaaS Compliance Dashboard

Annual subscription for TNPCB / municipal bodies to view real-time compliance data across units.

03

AMC — Maintenance

Annual contract for sensor recalibration, probe cleaning, and tamper-audit servicing.

04

Regulatory Partnership

Position as mandated compliance infrastructure via TNPCB — pull-through demand, not opt-in sales.

FINANCIALS

Core Sensing Stack

Turbidity sensor module	₹300 – 560
pH sensor + probe	₹1,200 – 1,550
TDS sensor module	₹480 – 1,000
TCS3200 color sensor	₹350 – 450
ESP32 dev board	₹450 – 600
Wiring, enclosure, misc.	₹300 – 500

Shared by both plans: ₹3,100 – 4,600

Plan A — Detect + Divert

Solenoid valve (1/2")	₹220 – 700
Relay / MOSFET driver	₹60 – 150
12V power adapter	₹200 – 350
Flyback diode	₹5 – 10

Total: ~₹3,600 – 5,800

Plan B — + Treat

Holding tank (5–10L)	₹150 – 400
Overflow float sensor	₹100 – 200
Dosing pump + coagulant	₹700 – 1,100
2nd relay channel	₹60 – 100

Total: ~₹4,600 – 7,600

Prototype vs. commercial: figures above are prototype cost. Industrial-grade probes, enclosure, valves, and certifications push a deployable unit to ₹40,000 – ₹1,00,000.

Calibration & drift: the AMC (slide 10) covers weekly probe calibration against a reference sample and periodic TCS3200 white-balance recalibration to control sensor drift.

WHO THIS IS BUILT FOR

**TARGET
AUDIENCE**

POLLUTION CONTROL BOARDS

TNPCB & municipal bodies needing auditable, real-time compliance data instead of periodic inspection.

DYEING UNIT OWNERS

Small & mid-sized units wanting a low-cost way to prove compliance and avoid penalties or shutdowns.

CETP OPERATORS

Common Effluent Treatment Plants that need per-source visibility into which unit is sending non-compliant batches.

ENVIRONMENTAL NGOS & RESEARCHERS

Groups tracking river health who need continuous, verifiable discharge data rather than spot samples.

AN HONEST READ OF WHERE THIS STANDS

SWOT ANALYSIS

✓ Strength

- Detect + act in one device not just monitoring
- Directly matches a real, local regulatory problem
- Modular — Plan A cheap entry, Plan B full pre-treatment

⚠ Weakness

- Rule-based classifier needs real effluent data to become a trained ML model
- Actuation hardware raises cost vs. sensor-only competitors
- Requires field validation before regulator trust is earned

🔒 Threat

- Factories may attempt to bypass or tamper with hardware
- Established water-quality sensor brands as competitors
- Slow regulatory adoption cycles

📈 Opportunity

- TNPCB mandate could make this required infrastructure
- Scalable to other pollution-heavy industrial clusters
- Potential CSR / green-tech grant funding



WHY IT MATTERS BEYOND THE PIPE

IMPACT & BENEFITS

Environmental

- Reduces untreated dye discharge into rivers & groundwater
- Cuts TDS / chloride load reaching sensitive downstream zones
- Supports long-term river and farmland recovery

Economic

- Avoids penalties & shutdown risk for compliant units
- Lower-cost alternative to full-scale ETP upgrades
- New compliance-tech revenue stream for the region

Social

- Protects groundwater used by nearby farming communities
- Builds public trust in industrial accountability
- Creates a replicable model for other pollution clusters

FOUR PHASES TO STATE-WIDE SCALE

**FUTURE
IMPLEMENTATION**

1

Phase 1 — Prototype & Calibrate

Build Plan A on a single outflow line; collect labeled samples from a partner dyeing unit and establish the weekly calibration routine.



2

Phase 2 — Pilot at One CETP

Deploy on 2–3 member units feeding a single CETP in Erode / Tirupur; validate diversion accuracy and begin training the ML classifier on field data.



3

Phase 3 — Add Treatment (Plan B)

Introduce the holding-tank + dosing layer once diversion logic is proven reliable in the field.



4

Phase 4 — Regulatory Partnership & Scale

Approach TNPCB to position ChromaGuard as recommended compliance infrastructure across the dyeing belt.

Long-term vision: a state-wide effluent interception network, feeding a live compliance map for TNPCB across every textile cluster in Tamil Nadu.

THANK YOU

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